

Modeling Bubble Flow and Current Density Distribution in an Alkaline Electrolysis Cell

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ABSTRACT

The effect of bubbles on the current density distribution over the electrodes of an alkaline electrolyzer cell is studied using a two-dimensional computational fluid dynamics model. Model includes Eulerian-Eulerian two-phase flow methodology to model the multiphase flow of Hydrogen and Oxygen with water and the behavior of each phase is accounted for using first principle. Hydrogen/Oxygen evolution, flow field and current density distribution are incorporated in the model to account for the complicated physics involved in the process. Fluent 6.2 is used to solve two-phase flow and electrochemistry is incorporated using UDF (User Defined Function) feature of Fluent. Model is validated with mesh refinement study and by comparison with experimental measurements. Model is found to replicate the effect of cell voltage and inter-electrode gap (distance between the electrodes) on current density accurately. Further, model is found to capture the existence of optimum cell height. The validated model is expected to be a very useful tool in the design and optimization of alkaline electrolyzer cells.

Keywords: Alkaline Electrolysis, Hydrogen production, Two-Phase Flow, Optimum Cell Height, Ion transport

1. INTRODUCTION

Alkaline electrolysis is an attractive path toward renewable Hydrogen production. However, cost is a big impediment to using hydrogen more widely as a fuel. Low cost alkaline electrolysis requires innovation in cell and system design. Computational fluid dynamics (CFD) model presented in this paper is developed to aid in the design and optimization of such electrolyzer cells.

The presence of gas phase in the liquid electrolyte of the alkaline electrolyzer makes it a two-phase flow problem. Bubbles nucleate and grow on electrode surfaces due to electrochemical reaction. During their growth, bubbles make part of electrode inactive for reaction. Once bubbles reach bulk electrolyte flow, they decrease the effective electric conductivity of the electrolyte by reducing the cross-sectional area of pure electrolyte available for current transport. Hence, bubble/gas release and motion significantly affect the performance of the alkaline electrolyzer.

Dahlkild[1] applied the mixture two-phase flow model for electrolysis process and implemented a boundary layer analysis. The numerical results showed that ionic species concentrations are essentially homogeneous as the mixing effect of the bubble suspension usually is much larger than dispersion by molecular diffusion even at laminar flow conditions. He also showed that the non-uniformity of the bubble distribution along the electrode results in a non-uniform current density distribution. Mat et al.[2,3] developed a two-phase mathematical model for hydrogen production in a forced flow electrochemical system and found that model successfully captures the main characteristics of electrolysis process. Nagai et al.[4] experimentally investigated effects of current density and space between electrodes on efficiency of water electrolysis. They also studied the

effects of electrode heights, system temperature and electrode inclination in the presence as well as absence of separator.

In this paper, Eulerian-Eulerian two-phase flow model is applied and the behavior of each phase is accounted for using first principle. Mat et al. [2,3] also used a similar model to simulate the two-phase flow in the cathode channel (space between cathode and membrane) of an alkaline electrolysis cell. However, work reported in the current paper is for a complete electrolyzer cell *i.e.*, for the cathode channel as well as the anode channel. Motivation for developing such a model is to develop a modeling tool that will ultimately aid in the design of industrial alkaline electrolyzer cells. Such a model could also give insights into the electrolyzer physics outside the immediate range of experiments performed for validation studies. In the current study, commercially available Fluent 6.2 software is used to develop the computational fluid dynamics model for alkaline electrolysis. Hydrogen evolution, flow field and current density distribution are incorporated in the model. Electrochemistry is incorporated into the multiphase module of Fluent using the feature of User Defined Functions (UDF). Corresponding mathematical formulation is presented in the next section followed by the details of the experimental setup developed for generating results for validation of the model. Results of the model are found to be valid both qualitatively as well as quantitatively. This model can thus serve as a valuable tool in the design and optimization of alkaline electrolyzer cells.

2. MATHEMATICAL FORMULATION

2.1 Two-Phase Flow

Figure 1 shows a schematic sketch of the electrolysis cell considered in this paper. The cell consists of two electrodes, liquid electrolyte and a membrane (separator). Electrodes are aligned vertically to facilitate gas removal by density difference as well as by convection. They are assumed to be infinitely thin with electrochemical reactions occurring on their surfaces that are in contact with the electrolyte flow. Membrane (separator) is used in the electrolysis cell to separate hydrogen from mixing with oxygen.

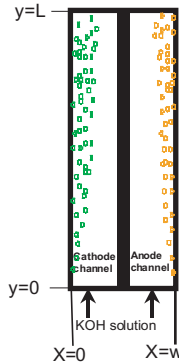


Figure 1. Schematic sketch of the electrolysis cell considered.

In order to simulate the flow behaviour in the system, different phases are treated mathematically as interpenetrating continua. Since the volume occupied by a phase cannot be occupied by the other phases, the concept of phase volume fraction is introduced. These volume fractions are assumed to be continuous functions of space and their sum is equal to one, *i.e.*,

$$\alpha_l + \alpha_g = 1, \quad (1)$$

where subscripts *l* and *g* refer to the liquid and gas phases, respectively. Conservation equations for each phase are derived to obtain a set of equations, which have similar structure for all phases. Constitutive relations that are obtained from empirical information close these equations.

Continuity:

$$\frac{\partial}{\partial t}(\alpha_i \rho_i) + \nabla \cdot (\alpha_i \rho_i \bar{v}) = 0 \quad (2)$$

Momentum:

$$\frac{\partial}{\partial t}(\alpha_i \rho_i \bar{v}) + \nabla \cdot (\alpha_i \rho_i \bar{v} \bar{v}) = -\alpha_i \nabla p + \nabla \cdot (\alpha_i \mu_{eff} \nabla \bar{v}) + \alpha_i \rho_i \bar{g} + \kappa_{ji} \bar{v}_r + \kappa_{ji} \left(\frac{D_j}{\sigma_{ji} \alpha_j} \nabla \alpha_j - \frac{D_i}{\sigma_{ji} \alpha_i} \nabla \alpha_i \right), \quad (3)$$

where subscripts i and j represents phases and take the value of l and g in this problem. Effective viscosity, μ_{eff} is a combination of laminar and turbulent contributions. The effective viscosity is calculated using the Realizable $k-\epsilon$ turbulence model embodied in the Fluent CFD package. $\alpha_i \rho_i \bar{g}$ in Eq. (3) is the buoyancy force, where $\bar{g}$ is the gravity vector. $\kappa_{ji} \bar{v}_r$ is the inter-phase friction term or the drag force between two phases, where $\bar{v}_r$ is the slip velocity vector between two phases and $\kappa_{ji} \left(\frac{D_j}{\sigma_{ji} \alpha_j} \nabla \alpha_j - \frac{D_i}{\sigma_{ji} \alpha_i} \nabla \alpha_i \right)$ is the momentum exchange due to volume fraction gradient driven inter-phase mass diffusion, where D is the diffusivity and σ is the dispersion Prandtl number.

The interphase exchange coefficient κ_{ji} in Eq. (3) can be written in the following general form:

$$\kappa_{ji} = \frac{\alpha_i \alpha_j \rho_j f}{\tau_j}, \quad (4)$$

where f , the drag function, is defined as

$$f = \frac{C_D \text{Re}}{24}, \quad (5)$$

$$\text{while,} \quad C_D = \begin{cases} 24(1 + 0.15 \text{Re}^{0.687}) / \text{Re}, & \text{Re} \leq 1000 \\ 0.44, & \text{Re} > 1000 \end{cases} \quad (6)$$

Re is a function of the bubble diameter (d). In the current study, bubble diameter is assumed to be constant value of 100 microns.

2.2 Ionic Species Transport and Potential Field

Since the mixing effect of the bubble suspension usually is much larger than dispersion by molecular diffusion even at laminar flow conditions, ionic species concentrations are assumed to be essentially homogeneous in this study. Therefore, the potential field is governed by

$$\nabla \cdot (\sigma^{eff} \nabla \varphi_i) = 0, \quad (7)$$

where σ^{eff} is the effective conductivity of the electrolyte and can be calculated using the Bruggeman correction as

$$\sigma^{eff} = \sigma_0 (1 - \alpha)^{1.5}. \quad (8)$$

Equation (7) is solved in Fluent using the User Defined Scalar feature of Fluent. The boundary conditions associated with Eq. (7) are

$$x = 0, i_c = -i_{0,c} (1 - \alpha_{H2}) \exp \left(-\frac{F}{2RT} \eta_c \right) \quad (9)$$

$$x = w, i_a = i_{0,a} (1 - \alpha_{O2}) \exp \left(-\frac{F}{2RT} \eta_a \right) \quad (10)$$

where i_0 is the exchange current density, $(1 - \alpha)$ represents the reduction of active electrode area due to the bubbles on the electrode and the overpotential η is defined by

$$\eta = \varphi_e - \varphi_i - U, \quad (11)$$

where ϕ_c is the electric potential and U is the open circuit potential. Various values of the numerical constants used in the computations are listed in Table 1.

The hydrodynamic and electrochemical components of the model are coupled through the source terms in the continuity equation and through the voidage of the electrolyte solution due to gas bubbles. Mass of Hydrogen produced at the cathode is obtained from the local current density value using the first law of electrolysis. This value of Hydrogen mass is given as the mass 'source' value in the continuity equation for Hydrogen. Similarly, mass of Oxygen produced at the anode is obtained from the local current density and given as mass 'source' value in the corresponding continuity equation. Mass 'sink' term for water is also obtained using the first law of electrolysis and added in it's continuity equation. Voidage computed with these source/sink terms is used to find the new local current density.

Table 1. Numerical values of constants used in computations

Symbol	Value
ρ_l	1050 kg/m ³
d	100e-6 m
σ_0	50 – 100 S/m
i_0	0.2 A/m ²
U	1.29 V
T	20 – 60 °C
F	96487 C/mol
R	8.314 J/mol

3. EXPERIMENTAL DETAILS

Current-cell voltage curves were recorded in a 5"x5" square parallel plate divided or undivided flow cell equipped with a Raney nickel cathode and 204 stainless steel anode. The data was obtained at temperatures ranging from 20 to 60°C, using a Xantrex XHR 20-50 power supply as a DC source. The electrocatalytic cathodes were prepared by co-depositing nickel and zinc or nickel and aluminum onto a stainless steel 204 substrate by a wire-arc deposition method. Most of the zinc or aluminum was then removed from the surface of the electrode by leaching in circulating 30% KOH solution at 60°C for periods of 6 to 24 hours. The anode material was 204 Stainless Steel. The active electrode surface area was 161cm². 30% w/w KOH solution was prepared by dilution of Fisher Certified 45% w/w solution with de-ionized water.

In a typical experiment, a 30 % KOH solution at temperatures of 20 to 60°C was circulated through the cell at flow rates of up to 2L/minute. The current was increased in 5A increments, and the cell voltage at each current was recorded manually. Typical results from these experiments are shown in the graph below.

4. RESULTS AND DISCUSSION

Physics of alkaline electrolysis is very complicated as it involves multiphase flow, electrochemical reactions and ion transport. Therefore, the model is set up over a simple two-dimensional (2D) domain. Physical validity of such 2D models can be quickly checked. Further, experimental validation of such a 2D model can be verified easily by performing simple experiments at a small scale. Once built and validated, extending the model to the third dimension would not be a difficult exercise.

Mesh independence of the code is verified using 30×127 and 75×250 elements. Results obtained with the two meshes (not presented here for brevity) are found to be essentially the same. Run with finer mesh is however, five times more computationally expensive than the run on the coarse one. Therefore, for computational economy, all subsequent runs are performed on the coarser mesh. The average current density predictions of the model are compared with experimental results generated over a single-cell setup that is described in the previous section. The simulations are compared with experiments in Figure 2. It can be observed that the predictions are in good comparison with experimental observations over the considered range of cell heights and cell voltages. This establishes the experimental validity of the model.

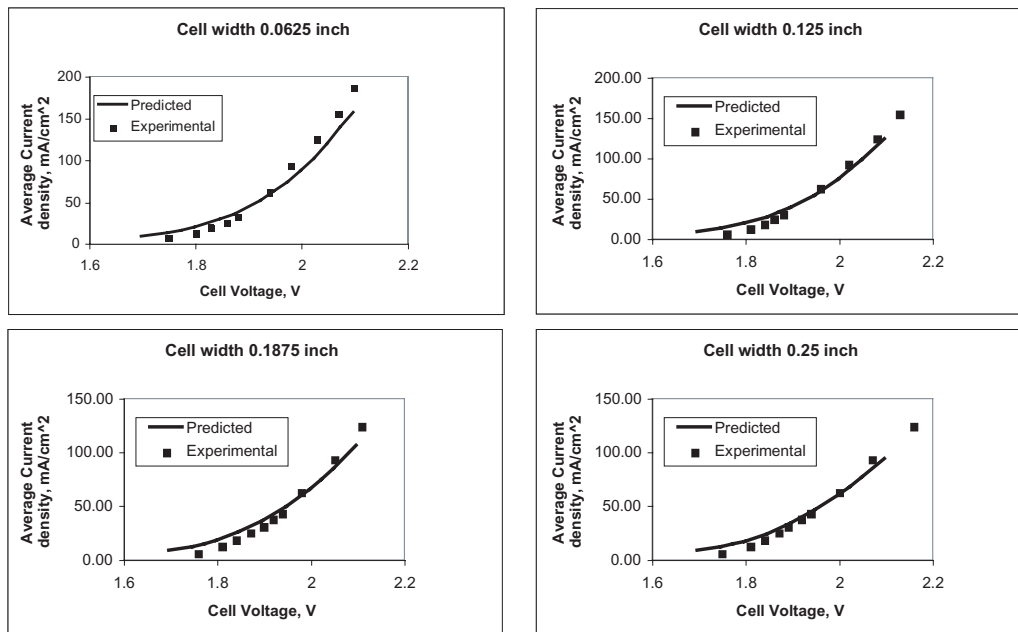
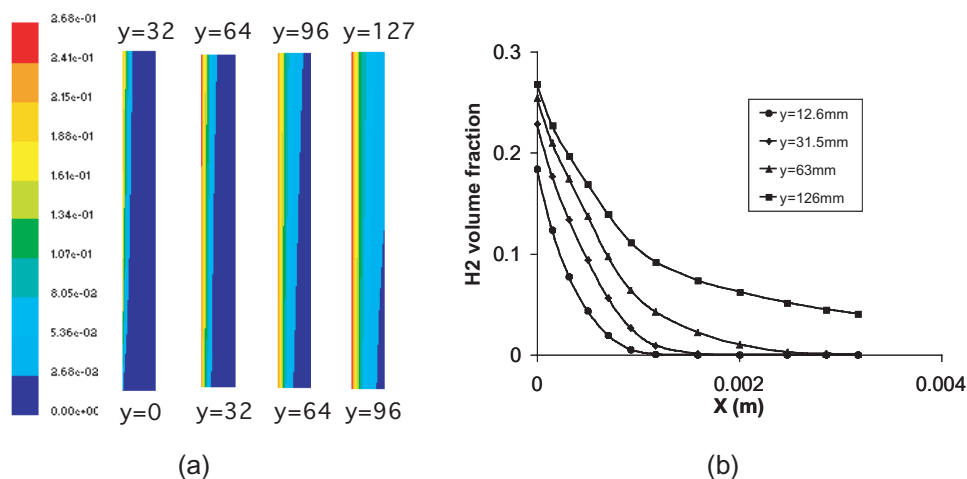


Figure 2. Comparison of model predictions with experimental observations

Figure 3(a) shows the spread of H_2 in several vertical sections of a 127mm electrode channel and Figure 3(b) shows H_2 volume fraction profiles at several 'y=constant' lines of the electrode channel. As expected, H_2 volume fraction in the electrode channel increases as we move in +y direction due to the diffusion of H_2 in the horizontal direction. Figure 4 shows H_2 volume fraction and current density variations along the cathode. Gas accumulation increases along the electrode due to the convective effect of the electrolyte flowing in +y direction. This increase in gas accumulation results in a gradual decrease of current density along the electrode. These intuitively appealing predictions are in line with Dahlkild[1], Mat et al[3] and Mat and Aldas[2].

Nagai et al[4] reported an optimum inter-electrode gap of 1-2 mm in their alkaline electrolyzer of stagnant electrolyte. Optimum inter-electrode gap can be expected to be much less in the case of moving electrolyte. Further, this optimum inter-electrode gap decreases significantly with increase in the electrolyte flow rate.

Figure 3. Variations of H_2 volume fraction in the electrode channel

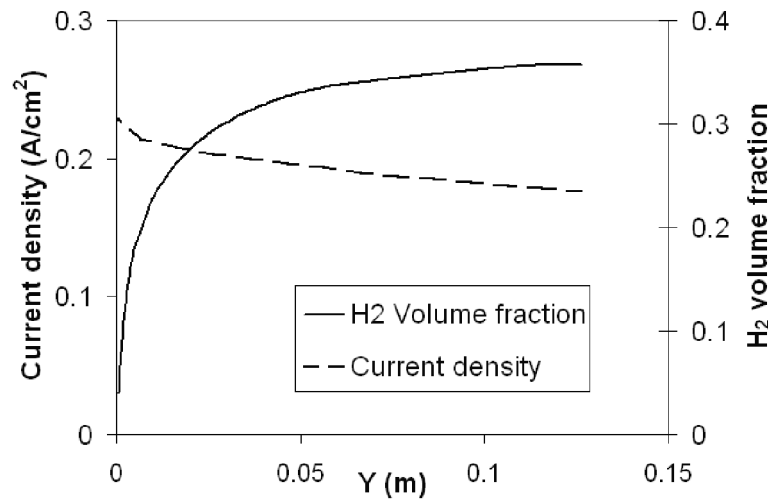


Figure 4. Variations of current density and H2 volume fraction along the electrode

As depicted in Figure 5, this is captured very well by the current CFD model. At a particular inlet velocity, average current density increases with decrease in inter-electrode gap up to a certain threshold inter-electrode gap only. This increase could be attributed to the decrease in the resistance to the flow of ions. Decrease in resistance to the flow of ions is due to decrease in the distance to be travelled by ions to reach the anode from cathode. The average current density decreases beyond the threshold inter-electrode gap due to severe accumulation of gases. Accumulation of gases significantly covers the electrode active area and also hinders the flow of ions from cathode to anode. It is also evident from Figure 5 that the threshold inter-electrode gap decreases with increase in flow rate. Decrease in the threshold inter-electrode gap with increase in flow rate could be attributed to the greater convective effect on the gaseous phases at higher flow rate.

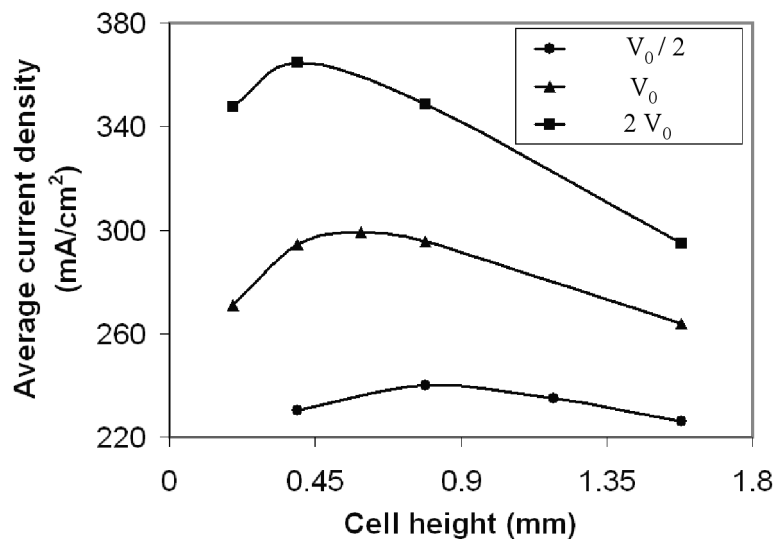


Figure 5. Variation of average current density with inter-electrode gap at a particular inlet velocity. Symbols are model predictions and lines are for illustration purpose only

Authors used this validated model to study the effect of electrode shape on the performance of an alkaline electrolyzer cell. This effort resulted in a novel electrode and a US patent application (Ravichandra et al[5]) is filed on this electrode. A summary of this study will be the subject of a subsequent paper. Further, the alkaline electrolyzer designed with inputs from this model has been awarded the Innovation Breakthrough Award for 2006 by the US based Popular Mechanics Magazine.

Authors also developed a 3D model in similar lines and a paper on its development and results is in preparation for submission elsewhere. The 3D model would serve as a very valuable tool in the design and optimization of alkaline electrolyzer cells. Authors plan also includes extension of the 3D cell model to an alkaline electrolyzer cell stack by coupling the CFD model with their model (Ravichandra et al[6]) for the prediction of shunt currents

5. CONCLUSIONS

Eulerian-Eulerian two-phase flow model available in Fluent 6.2 is used to develop a computational fluid dynamics model for multiphase flow involved in alkaline electrolysis. Electrochemistry is incorporated using the UDF (User Defined Function) feature of Fluent. Electrolyzer physics incorporated in the model is validated by mesh refinement study and by comparison with in-house experimental data.

Model predicts the existence of an optimum inter-electrode gap at any particular electrolyte flow rate. Further, optimum inter-electrode gap is found to decrease with increase in flow rate. Model would serve as a very useful tool in the design and optimization of alkaline electrolyzer cell and stack.

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